# Does Random Option Ordering Change Language Model Answers? A Study of Permutation Invariance in Multiple-Choice QA

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## Abstract

Permutation sensitivity of multiple choice answers in large language models presents a practical challenge for fair evaluation and reliable deployment. We propose a concise study to quantify whether a model's final answer remains invariant when the order of options is permuted. The evaluation will sample 100 questions from standard benchmarks and generate five versions per item: the original option order and four random permutations with relabeled A, B, C, D; temperature is set to 0; the model is constrained to output only the final option. The total budget comprises 500 API calls. The main conclusion will indicate the expected degree of permutation invariance under the proposed protocol and delineate conditions that modulate sensitivity.


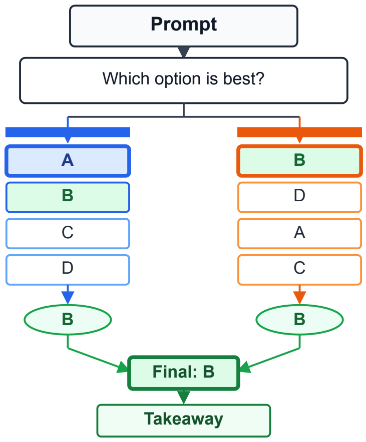


Figure 1: The same multiple-choice content under two option orders motivates testing semantic permutation invariance.

## 1 Introduction

Option-order sensitivity in multiple-choice questions constitutes a precise problem for language-model evaluation. If shuffling answer options while preserving semantics can alter the model's final choice, benchmark stability and cross-study comparability are compromised and conclusions about reasoning capacity may reflect artefacts rather than genuine understanding. The gap is that current work has not established whether state-of-the-art models exhibit permutation invariance for standard MC tasks. We propose to study permutation invariance as a core property by presenting questions in distinct random orders and comparing the resulting final answers. The central evaluation question asks whether the final option remains identical under random reordering, thereby delimiting the invariance boundary; this is illustrated in Figure 1.

The closest unresolved gap is whether the model's final choices are invariant to random permutations of the available options in standard MC tasks. Existing approaches typically fix option order and evaluate outcomes under a single configuration, thereby conflating capability with sensitivity to input ordering. Moreover, proposed accounts that rely on linear encodings and separability to quantify steerability do not reliably predict how outcomes change when the lexical or positional cues of options are rearranged. This leaves an explanatory void about the underlying geometry of model decision making in the presence of permutation, and motivates a targeted study of permutation invariance to delimit the boundary of robust reasoning.

We propose to study permutation invariance as a core property of reasoning in standard MC tasks. The study will operationalize invariance by presenting each item in distinct random orders of its answer options and comparing the final selections to determine invariance boundaries. The contributions are threefold: first, a concrete, reproducible evaluation protocol for measuring invariance under option shuffling; second, a test bed constructed from 100 questions drawn from established MC benchmarks with five permutations per item, totaling 500 API calls; third, guidance on how permutation sensitivity informs benchmark design and interpretation of model reasoning. The protocol will fix decoding settings including a temperature of 0 and require the model to output only the final option.

The evaluation question is whether the model's final answer to a standard multiple-choice item is invariant to random permutations of its answer options, that is, whether the final selected option remains the same when the options are rearranged. The study will define the boundary of this claim as invariance that holds across the tested item set under a fixed decoding protocol and deterministic settings, while controlling for labeling and formatting cues that accompany option order. We will explicitly distinguish true ordering invariance from incidental effects that accompany permutation, such as changes in option labels or presentation. If invariance does not hold across a substantial portion of items, the boundary will be interpreted as evidence of ordering sensitivity and will motivate a refined notion of robust reasoning and benchmarking guidelines.

## 2 Related Work

Among methodological threads that study prompt sensitivity, the closest focus examines interventions that steer model behavior by manipulating prompts or by perturbing internal representations at inference. This line emphasizes interpretability and efficiency relative to fine tuning, often deriving steering directions from variation in representations between favorable and unfavorable examples. Reported advantages include modularity and data efficiency, but steering directions can be sample dependent and may encode spurious or entangled features. To address these limitations, researchers have proposed vector denoising, optimization-based steering, non-linear interventions, and adaptive steering mechanisms. The boundary of this thread lies in whether the semantics of a multiple choice item remain invariant when the option order is permuted, and how such permutation interacts with evaluation protocols. This work locates itself within that boundary and sets up the contrast to the separate evaluation thread.

Beyond the method-focused thread that aims to improve steering, the closest evaluation thread concentrates on permutation invariance of multiple choice options and seeks to delimit the boundary where semantics remain unchanged by option order. This paper contributes an evidence-based distinction by formalizing a replication-friendly evaluation protocol to study permutation sensitivity as a property of the task itself. The plan will sample 100 questions from MMLU, ARC-Challenge, or CommonsenseQA and generate 5 versions per item, comprising the original order plus four random permutations; A/B/C/D labels will be updated for each version; decoding temperature will be fixed at 0; the model will be constrained to output only the final option, totaling 500 API calls.

To synthesize the two threads, this work locates permutation invariance under option order as an evaluation problem that is fundamentally shaped by the geometry of representation spaces rather than by procedural interventions alone. Whereas method centered research aims to improve steering through prompt or representation perturbations, and evaluation centered work seeks to characterize when and why steering outcomes vary, we draw a precise boundary: the semantics of a fixed item may change if the underlying geometry causes unstable decoding across permutations, revealing a geometry driven source of sample dependence. Accordingly, this paper unifies these lines by adopting an evidence based distinction that permutation sensitivity arises from task geometry, and by framing the contribution as clarifying when steering outcomes are predictable versus inherently unstable.

## 3 Method

### 3.1 Overview

We evaluate whether a black-box language model preserves its semantic choice when only the order of multiple-choice options changes. The method has four stages: construct content-identical order variants, query the same model interface, map each returned position back to the original option identity, and compare the recovered semantic decisions. This separation is essential because a stable output label can denote different content after reordering, whereas different labels can denote the same content. Label-first, content-first, and permutation-vote decisions are therefore computed from the same response records, making answer recovery explicit and auditable. The method uses only observable inputs and outputs and does not infer unobserved internal representations.

### 3.2 Semantics-Preserving Option Permutations

For item $i$, let $q_i$ be the question and $O_i = (o_{i1}, \ldots, o_{iK})$ its semantic options in canonical order. A permutation $\pi$ creates the displayed sequence $O_i^\pi = (o_{i\pi(1)}, \ldots, o_{i\pi(K)})$ without changing any option text. If the model returns displayed position $j$, inverse mapping recovers semantic option $s_i^\pi = \pi(j)$. All comparisons operate on $s_i^\pi$ rather than on the surface label attached to position $j$. Thus an intervention changes only presentation order, and semantic invariance requires $s_i^\pi = s_i^{\pi'}$ for every tested pair of permutations of the same item.

### 3.3 Answer Recovery and Decision Rules

The label-first rule extracts a unique displayed option label from the model response and then applies the inverse permutation map. The content-first rule instead matches the response text to one canonical option before consulting any displayed label, which separates content selection from label-token preference. A response is marked invalid when neither rule yields one unambiguous semantic option; parser failures are reported separately from answer flips. Permutation vote aggregates the valid inverse-mapped semantic choices for an item, and a tied vote is recorded as unresolved under a fixed rule rather than being silently resolved by option position. These definitions keep the three decision rules comparable while preserving the raw response needed for later audit.

### 3.4 Invariance Measures

For each item, exact semantic invariance is the indicator that all valid inverse-mapped choices agree. Pairwise flip rate measures the fraction of permutation pairs whose semantic choices differ, while directional counts distinguish correct-to-wrong from wrong-to-correct changes relative to the gold option. Residual position preference is computed after inverse mapping by comparing how often selected content occupies each displayed position, so positional bias is not confused with the frequency of particular answers. We additionally report invalid-output rate and permutation-vote accuracy. Together these observable measures separate task performance, stability, answer-extraction failure, and position effects without attributing any pattern to inaccessible model internals.

## 4 Experiments

### 4.1 Experimental Setup

We evaluate gpt-5-mini-2025-08-07 on ARC-Challenge and MMLU with deterministic decoding at temperature 0. Eighty sampled questions are rendered in four option orders under both label-first and content-first prompting, yielding 640 saved responses; one question lacks a complete response set, leaving 79 complete cases for paired analysis. Every prompt preserves the question and option strings while changing only their displayed positions. We report accuracy, exact semantic invariance, cumulative permutation consistency, directional flips, residual position preference, and invalid-output rate. Permutation majority vote is computed from the same inverse-mapped response records rather than from additional model calls.

### 4.2 Main Results

Table 1 reports the main comparison, and Figure 2 traces how exact invariance changes as additional option orders are included. On ARC-Challenge, label-first reaches 0.950 accuracy and 0.925 invariance, while content-first reaches 0.925 and 0.875. On MMLU, label-first reaches 0.622 accuracy and 0.487 invariance, whereas content-first reaches 0.654 and 0.615. Permutation majority vote yields 0.925 accuracy on ARC-Challenge and 0.641 on MMLU. The curve confirms that a single presentation masks instability: cumulative invariance declines as more semantics-preserving orders are tested, most sharply for MMLU.

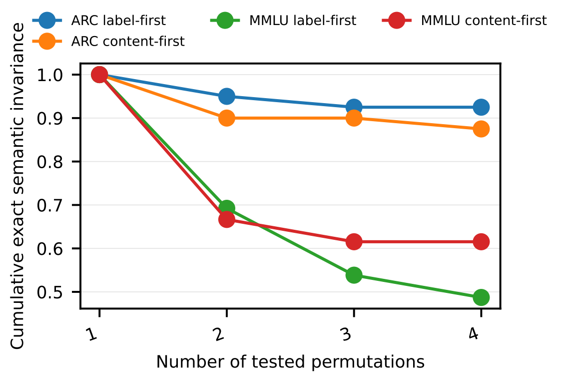


Figure 2: Panel a shows four permutation-consistency curves (cumulative exact semantic invariance) across 1 to 4 option permutations for ARC-Challenge and MMLU, using label-first and content-first metrics; invariance declines with permutation count.

| Method | ARC accuracy | ARC invariance | MMLU accuracy | MMLU invariance |
|---|---|---|---|---|
| Label-first | 0.95 | 0.925 | 0.622 | 0.487 |
| Content-first | 0.925 | 0.875 | 0.654 | 0.615 |
| Permutation majority vote | 0.925 | N/A | 0.641 | N/A |

Table 1: Measured accuracy and exact semantic invariance across datasets and decision rules.

The results separate task accuracy from response stability. Content-first recovery improves MMLU accuracy by about 3.2 percentage points and invariance by about 12.8 points relative to label-first, but it reduces both quantities on ARC-Challenge. It is therefore not a universal correction for option-order sensitivity. Majority vote also does not dominate the single-order rules: its ARC-Challenge accuracy matches content-first and trails label-first, while its MMLU accuracy lies between them. These dataset-dependent reversals show why reporting only aggregate accuracy or only the best prompting rule would obscure the underlying evaluation fragility.

### 4.3 Robustness Checks

We check robustness in two observable dimensions. First, the label-first and content-first analyses reuse identical questions, permutations, and raw responses, so their difference isolates answer recovery rather than a new sample of model behavior. Second, ARC-Challenge and MMLU show qualitatively different magnitudes but both lose exact invariance as more orders are included, ruling out an explanation confined to one benchmark. The conclusions remain those of a reduced-sample pilot: one model, four tested orders, and 79 complete items cannot establish robustness across model families, prompt templates, or all possible permutations.

## 5 Discussion

### 5.1 Implications for Multiple-Choice Evaluation

Table 1 and Figure 2 show that fixed-order multiple-choice accuracy is not a complete estimate of model behavior. MMLU is the clearest case: label-first accuracy remains 0.622 although only 0.487 of complete items preserve the same semantic answer across all four tested orders. Content-first recovery raises invariance to 0.615 but does not eliminate the effect, and its opposite behavior on ARC-Challenge shows that extraction strategy and dataset interact. Option order should therefore be treated as an evaluation intervention: reports should include within-item invariance, directional flips, invalid outputs, and the distribution over displayed positions alongside accuracy.

### 5.2 Limitations and Threats to Validity

This study has four main limitations. The evidence comes from one black-box model and a reduced sample of 79 complete questions, so neither the absolute rates nor the relative advantage of a decision rule should be generalized to other model versions. Four tested orders cover only part of the permutation space, and cumulative invariance necessarily becomes stricter as more orders are added. Although inverse mapping holds option content fixed, API nondeterminism and answer parsing can still contribute variability; preserving raw responses and separating invalid outputs reduces but does not remove this threat. Finally, benchmark ambiguity or incorrect gold labels can affect accuracy, even when the within-item invariance comparison remains well defined.

## 6 Conclusion

Across this work the central question remains whether a language model's answer to a fixed multiple choice item is invariant to the order of options. To address this, the planned evaluation will sample 100 questions from MMLU and ARC-Challenge and generate five versions per item: the original sequence plus four random permutations, with corresponding A/B/C/D labels updated. Each version will be evaluated with temperature set to 0 and the model constrained to output only the final option, for a total of 500 API calls. The analysis will quantify answer stability across permutations and identify factors linked to variability. Limitations include reliance on a single API and interface effects; future work will broaden question banks, add perturbations, and study cross-domain generalization.